

Sapna Naga

Lead AI Engineer

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Experienced Lead AI Engineer with 8+ years of experience in Large Language Models (LLMs), Generative AI, predictive analytics, and AI automation across LegalTech, Healthcare, Insurance, CivicTech / GovTech, and Supply Chain domains. Proven track record of designing and deploying AI agents, LLM-powered applications, and predictive models that automate complex business workflows and enable intelligent data driven decision making. Skilled in developing AI solutions, building scalable data pipelines, integrating enterprise systems with LLM APIs, and deploying cloud-based AI applications to deliver high-impact Machine Learning systems that improve operational efficiency and drive multi-million-dollar business outcomes. Passionate about leveraging AI and data science to solve complex real-world problems, enhance decision making, and build scalable intelligent systems while mentoring teams to deliver production ready AI solutions.

Work Experience

JANUARY 2024 – PRESENT

Lead AI Engineer/LegalMente AI, Bengaluru, India

- Revamped AI Paralegal Assistant into a custom AI Agent leveraging multiple Large Language Models (LLMs) to handle complex legal and business tasks, including contract drafting and review, legal and general inquiries, startup advisory, ad and social media content creation, and sourcing relevant web materials.
- Designed and deployed an AI Paralegal Assistant capable of responding to legal and general inquiries with high accuracy and contextual understanding.
- Implemented an AI automation system for U.S. Business Entity Formation, optimizing the LLC and corporation setup process for efficiency and compliance.
- Developed an AI automated contract review system, integrating Generative AI and custom NLP models to analyze and extract key insights from U.S. contracts, including BAAs, SAFEs, and NDAs.
- Designed and built an AI Browser Agent leveraging browser-use.com and custom AI / NLP pipelines to autonomously navigate U.S. government websites and extract prior quarter ending balances for all current State Legislature members across both Houses and Senates.
- Developed a natural language AI agent using Voiceflow for conversational orchestration and n8n for ETL / data workflows to enable conversational querying of Monday.com, allowing users to retrieve stakeholder insights across boards and dashboards through a multi-modal interface.
- Built an AI agent using Voiceflow and n8n powered data pipelines to retrieve and reason over historical Fathom meeting transcripts and notes, enabling users to query past meetings and extract contextual insights.
- Delivered an MVP AI solution that processes scanned petition PDFs, extracts signer names and addresses, validates them against voter files, and generates CSV outputs with match results and confidence scores through a single command execution.
- Conducted technical and AI architecture assessment of a federal acquisition modernization proposal, evaluating Appian GAM based architecture, AI automation feasibility, system integration risks, and governance alignment against U.S. Coast Guard acquisition lifecycle requirements.

JANUARY 2023 – OCTOBER 2023

Associate Manager (Lead Machine Learning Engineer)/Factspan Analytics Pvt. Ltd., Bengaluru, India

- Developed and deployed a time-series forecasting model for Capacity Management in hospital admissions, discharges, and transfers (ADT), enabling accurate volume predictions across nurse units. Designed an interactive Tableau dashboard to visualize forecasts, empowering hospital operations teams to make proactive staffing decisions and eliminate bottlenecks – leading to \$10 million in savings over seven years.
- Mentored and guided junior team members in the design, optimization, and deployment of healthcare AI solutions, including, Pharmacy Refill Request Prediction (Improved acceptance and rejection probability analysis), Pharmacy Refill Volume Forecasting (Enhanced demand prediction for better inventory planning), Surgery Turnaround Time Prediction (Streamlined operating room scheduling, increasing efficiency), Radiology Report

Summarization & Care Coordination (Automated identification of critical findings using RAG to improve care workflows).

- Led Machine Learning and Python training programs for MTech, MBA graduates and junior team members, delivering hands-on sessions to strengthen expertise in advanced ML methodologies, ensuring team skill development and project readiness.
- Contributed to talent acquisition efforts by interviewing and evaluating candidates for data scientist and data analyst roles, ensuring the selection of top talent aligned with organizational goals.

MARCH 2022 – JANUARY 2023

Senior Analyst (Senior Machine Learning Engineer)/Factspan Analytics Pvt. Ltd., Bengaluru, India

- Developed a predictive analytics model for Emergency Department (ED) patient volume, forecasting short-term (72 hours) and mid-term (7 days) trends across multiple nurse units. Designed and implemented a Tableau dashboard integrating real-time and forecasted patient data, empowering hospital operations teams to optimize staffing and resource allocation. This initiative improved patient flow, reduced capacity bottlenecks, and decreased Imaging department's wait time by 20%, enhancing overall patient care.
- Mentored junior team members in machine learning, data analysis, and visualization, providing guidance on model development and best practices. Fostered a collaborative learning environment, accelerating their technical growth and contribution to key projects.

SEPTEMBER 2021 – FEBRUARY 2022

Info Technology Analyst (Data Science)/EAM/NTT DATA GDS, Bengaluru, India

- Extract, clean, and analyze complex datasets to identify trends and patterns, transforming raw data into actionable insights through advanced visualization techniques for effective client communication.
- Conduct statistical analysis and optimize machine learning pipelines by fine-tuning models based on test results, ensuring enhanced accuracy, efficiency, and overall performance.
- Design and develop interactive dashboards and reports, enabling stakeholders to access critical patient insights within seconds, leading to improved operational efficiency, enhanced customer experience, and cost savings.
- Collaborate cross-functionally with data scientists, business analysts, and stakeholders to align data-driven solutions with business objectives, ensuring scalability, security, and long-term impact.

MARCH 2020 – SEPTEMBER 2021

Software Development Analyst (Data Science)/Enterprise Analytics/NTT DATA GDS, Bengaluru, India

- Designed and developed predictive models using advanced machine learning algorithms to optimize healthcare outcomes and improve model performance.
- Conducted comprehensive Exploratory Data Analysis (EDA) and implemented Feature Engineering techniques to enhance data quality and model accuracy during the preprocessing phase, leveraging third-party datasets.
- Built an interactive dashboard that seamlessly integrates patient data from 13 disparate sources — including demographics, group coverage, benefits, and claims history—resulting in a 40% increase in employee productivity and a 15% reduction in operational costs.

DECEMBER 2019 – MARCH 2020

Software Development Senior Associate/Data Services/NTT DATA GDS, Bengaluru, India

- Played a pivotal role in implementing best practices for system enhancements, collaborating with cross-functional teams to assess critical information and successfully execute complex tasks.
- Analyzed and resolved intricate system issues by interpreting complex datasets and code, identifying root causes, and delivering articulate technical solutions to clients. Successfully implemented long-term fixes in production, reducing incident inflow by 40%.

JUNE 2019 – DECEMBER 2019

Software Development Senior Associate/Interfaces/NTT DATA GDS, Bengaluru, India

- Extract and manage data from multiple sources, ensuring accuracy, consistency, and seamless integration into internal databases and data warehouses while meeting SLA commitments.
- Optimized operational efficiency by automating data processing workflows using SSIS packages and SQL Server, reducing manual effort and increasing team productivity by 75%.

MARCH 2019 – JUNE 2019

Software Development Senior Associate/Insurance Web Application/NTT DATA GDS, Chennai, India

- Ensured accurate and up-to-date insurance product data for application users by managing product availability, associated funds, contract details, and resolving data-related issues.
- Developed and optimized ETL pipelines to extract, transform, and load data from multiple sources into Data Mart through nightly batch processes, enhancing data reliability and system performance.
- Proactively monitored application stability, identifying patterns in recurring failures and data inconsistencies, implementing permanent fixes to improve operational efficiency.

SEPTEMBER 2018– MARCH 2019

Software Development Senior Associate/Group Trading Platform/NTT DATA GDS, Chennai, India

- Automated the trading system to monitor fund allocations and transactions from the GTP of insurance companies, enhancing employee productivity by 60%.
- Optimized the data extraction process, reducing processing time by 70%, which significantly decreased overall load time and ensured updated data availability before the start of business hours.

MARCH 2018 – SEPTEMBER 2018

Software Development Senior Associate/CRM/NTT DATA GDS, Chennai, India

- Managed user access and role-based permissions for Microsoft Dynamics CRM, ensuring secure and efficient authorization while resolving customer enrollment issues, leading to improved service quality and user experience.
- Led incident management and stakeholder communication, collaborating with users, application owners, and senior leadership to swiftly address and resolve issues, ensuring minimal disruption to business operations.

OCTOBER 2017 – MARCH 2018

Software Development Sr. Associate/Insurance Electronic Doc Lib/NTT DATA GDS, Chennai, India

- Managed the deployment and maintenance of the SED Application, ensuring seamless upload and removal of electronic documents for over 500,000 policies.
- Designed and developed analytical reports to track application health, providing trend-based insights to optimize support service performance.

JUNE 2017 – OCTOBER 2017

Business Intelligence Trainee/Trainee/NTT DATA GDS, Hyderabad, India

- Trained on – DBMS/RDBMS, SQL, PL/SQL, DWH, ETL Process & Tools, SSIS, SSRS, Informatica.

Technical Skills and Expertise

AI and Machine Learning

- Expertise in AI Agents, Large Language Models (LLMs), and Generative AI, including building agentic AI systems that orchestrate multi-LLM workflows for legal document analysis, contract review, conversational querying, automated business workflows, and document intelligence tasks such as information extraction, semantic querying, and knowledge retrieval.
- Experience implementing Retrieval Augmented Generation (RAG) pipelines using vector embeddings, document retrieval, and LLM reasoning to deliver knowledge-grounded responses in production AI applications.
- Experience integrating multiple LLM providers and APIs (OpenAI, Grok, OpenRouter) to build production grade AI systems requiring contextual understanding, reasoning, and multi-step task execution.
- Advanced Time-Series modeling: multi-series clustering, PACF-based feature engineering, and predictive forecasting for healthcare operational analytics.
- Supervised and unsupervised machine learning: Regression, Classification, and Clustering algorithms for predictive modeling and pattern discovery.
- Machine Learning model development and Optimization: model training, hyperparameter tuning, and performance optimization for domain specific AI / ML systems.

Data Engineering and Visualization

- Strong proficiency in ETL processes, data integration, and large-scale dataset handling across distributed systems, including designing data pipelines that ingest structured and unstructured data sources (documents, transcripts, web data) for downstream AI and LLM based processing.
- Experience in preparing datasets and knowledge bases for semantic retrieval, AI search, and LLM-based applications.
- Expertise in feature engineering and Exploratory Data Analysis (EDA) to optimize machine learning model performance.
- SQL, MSSQL, and SSIS for robust database management, querying, optimization, and large-scale data processing.
- Data visualization expertise using Tableau, enabling insightful and data driven decision making.

Programming and Tools

- Python: Pandas, NumPy, Scikit-learn; experience integrating LLM APIs (Grok, OpenAI, OpenRouter) with enterprise applications through REST APIs and automation pipelines.
- Agentic tool orchestration and workflow automation: Voiceflow (AI agent interfaces), n8n (ETL and data workflows), and browser-use for autonomous browser-based agent interactions.
- Cloud Computing: Experience deploying AI and analytics solutions on AWS and Azure.
- Automation & Deployment: Expertise in data pipeline automation, MLOps, and containerized deployments, with experience deploying AI-powered services and automation workflows in production environments emphasizing scalability, reliability, and security.

Business Applications/Domains

- LegalTech
- Healthcare
- Insurance
- CivicTech / GovTech
- Supply Chain

Leadership and Mentorship

- Technical mentorship: Trained and guided ML engineers & data scientists in model development and deployment.
- Recruitment & Evaluation: Conducted technical assessments for hiring data science professionals.
- Cross-functional collaboration: Partnered with business stakeholders to align AI solutions with strategic goals.

Education

AUGUST 2012 – JULY 2016

Bachelor of Engineering in Electronics and Communication Engineering/SJB Institute of Technology

Certifications

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